University for the Creative Arts

BSBI School of Business and Innovation

Essay / Assignment Title: Development of an End-to-End NLP Pipeline for Chatbot Design Support

Programme Title: Natural Language Processing

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Year: 2026

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Introduction

Brief Context of the Project

Conversational AI is often used in customer support to manage customer inquiries efficiently and quickly. In banking, chatbots assist their customers in areas such as account management, card services, payments and transfers, and transaction issues. These systems make use of Natural Language Processing (NLP) to interpret the user's purpose and offer suitable responses. The real-world Banking77 dataset, comprising 13,069 queries in 77 intent categories, is used in this project. Various scenarios occur in the banking sector, such as card delivery, cash withdrawal, transfers, account problems, and more. It is also labelled, which serves as a good starting point to build and evaluate chatbot intent recognition models with state-of-the-art NLP techniques.

Project Objective

The goal of this project is to develop and evaluate an intent recognition end-to-end NLP pipeline for a banking chatbot. First, text will be preprocessed and perform exploratory analysis with NLTK and spaCy, exploring the data's language features will be done. Then, some basic models will be created by Logistic Regression, Naive Bayes and Linear Support Vector Machines with features as TF-IDF. Word2Vec embeddings will be explored, and then the use of a retrieval-based chatbot to give a deeper look at semantics will be demonstrated. In addition, a DistilBERT transformer model will be fine-tuned to classify intents and benchmark it with previous techniques. Finally, the responsible aspect of AI, model reliability, and deployment concerns are discussed, providing insights into the intricacies of implementing NLP-enabled chatbots in banking (Chuang and Cheng, 2022).

Methodology

An end-to-end NLP Pipeline has been developed for the Banking Chatbot Intent Identification with the Banking77 dataset. There were five stages in the process. Firstly, the text was preprocessed, and exploratory analysis was performed using NLTK and SpaCy for tokenisation, lemmatising, stop word removal and named entity recognition. Then, TF IDF features to train LR, NB, and Linear SVM have been created. Following that, Word2Vec embeddings were trained to learn about semantic relationships and demonstrate a retrieval Chatbot. A transformer model, DistilBERT, has been fine-tuned for intent classification purposes. Finally, the performance of the models has been analysed, the confidence scores have been examined, and ethical concerns have been explored in determining responsible deployment of chatbots (Kulkarni *et al*., 2023).

Task 1: Preprocessing & Exploratory Analysis for Chatbot Inputs

Dataset Inspection

The Banking77 dataset was picked for developing a banking chatbot intent recognition system. It has 9,993 training examples and 3,076 test examples, and customer queries are categorised into 77 intents that are related to banking. The few sample queries were related to several issues like delivery of cards, transfer of money, cash withdrawal and cash issue. The data appears to be ideal for use in testing conversational AI and intent classification methods (Ying and Thomas, 2022).

Dataset Link: https://huggingface.co/datasets/mteb/banking77

Exploratory Analysis

In Exploratory analysis, the distribution of intent was quite even, and relatively, there were about 158-187 examples in the most occurring class. This balance is good since it cuts down on model bias toward any single intent category (Allam *et al.*, 2025).

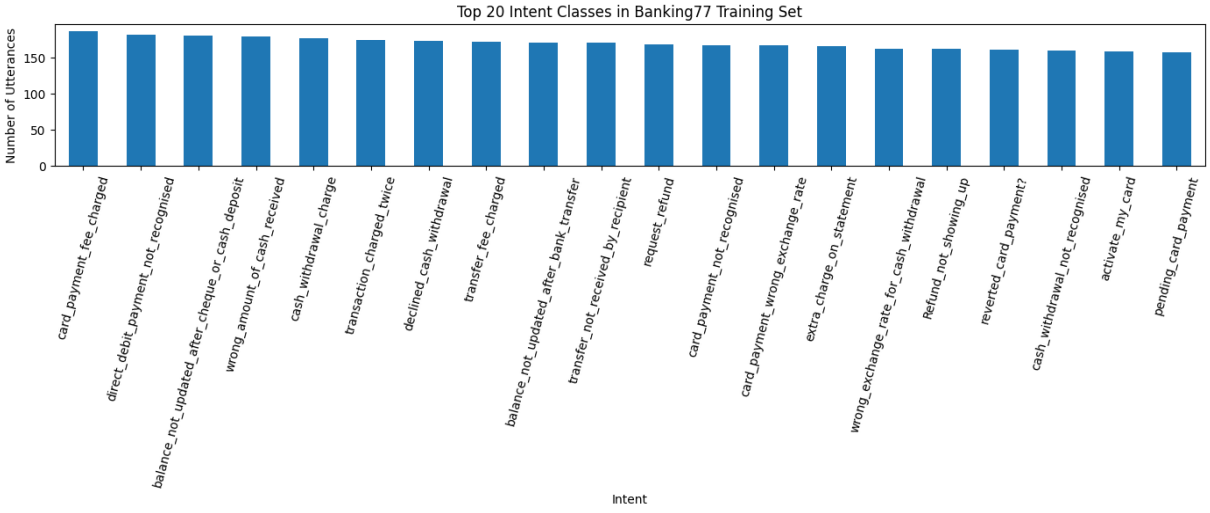


Figure 1: Intent Distribution Chart

(Source: self-developed)

Plus, text length analysis showed Banking77 consists of fairly brief customer queries averaging 11.96 words, with a median of 10. Most queries range from 5 to 15 words, which fits the succinct style of real chat sessions (Chang *et al.*, 2022).

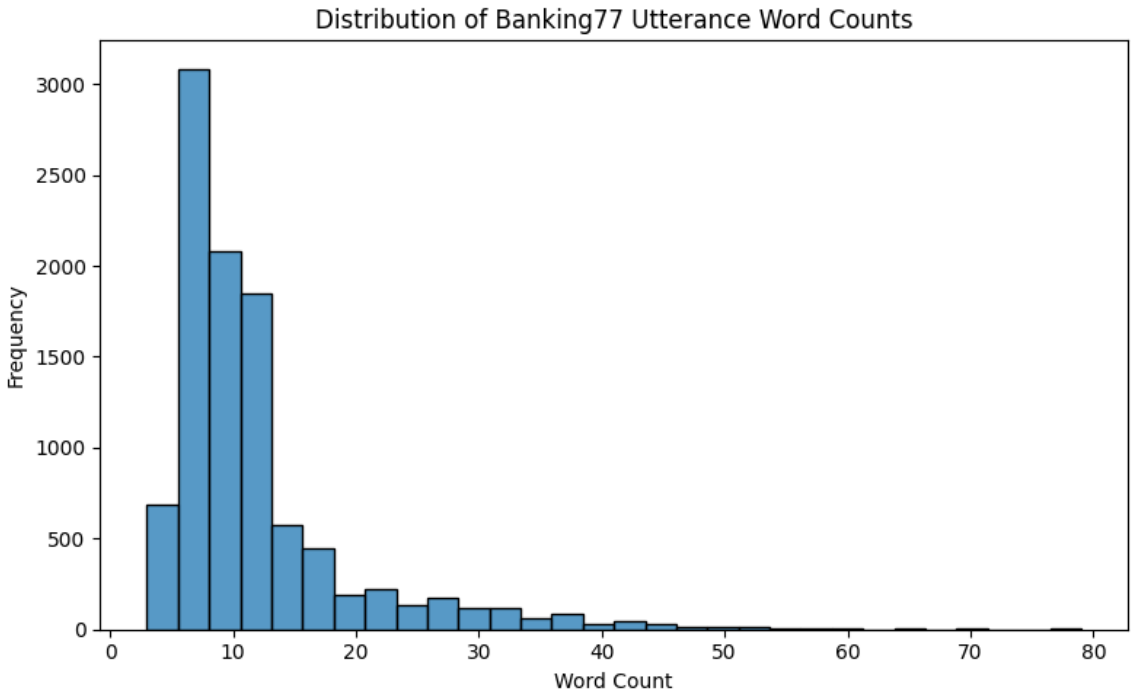


Figure 2: Word Count Distribution

(Source: self-developed)

Text Preprocessing

Both NLTK and spaCy were used for the tokenisation, removal of stopwords and lemmatisation. Think about it, when it has been asked, "Why has my new card still not come?" the question can be edited to only the essentials that have been conveying to the user. Overall, spaCy provided more succinct results, removing extraneous components and applying more stringent linguistic normalisation (Piasta and Kotas, 2025).

The average number of tokens for the NLTK is 5.66, and the vocabulary size is 393; for the spaCy, the average number of tokens is 4.53, and the vocabulary size is 291, as shown above. This means that SpaCy's behaviour when users type is more standardised, which may be helpful for other classification tasks that would be likely to be performed later.

Token and Entity Analysis

Then, the frequencies of the tokens were analysed, a step that came after a post-processing phase. Card, account, transfer, money and payment were found to be the most common banking terms and strengthened the link of the data with banking.

The frequencies of the tokens are presented in the following chart (Figure 3). Using spaCy's Named Entity Recognition, all types of info organisations, dates, locations, monetary amounts, and number expressions could be recognised. The most common entities were ORGs (Organisations), which were followed by DATE and GPE entities (Jang *et al.*, 2024).

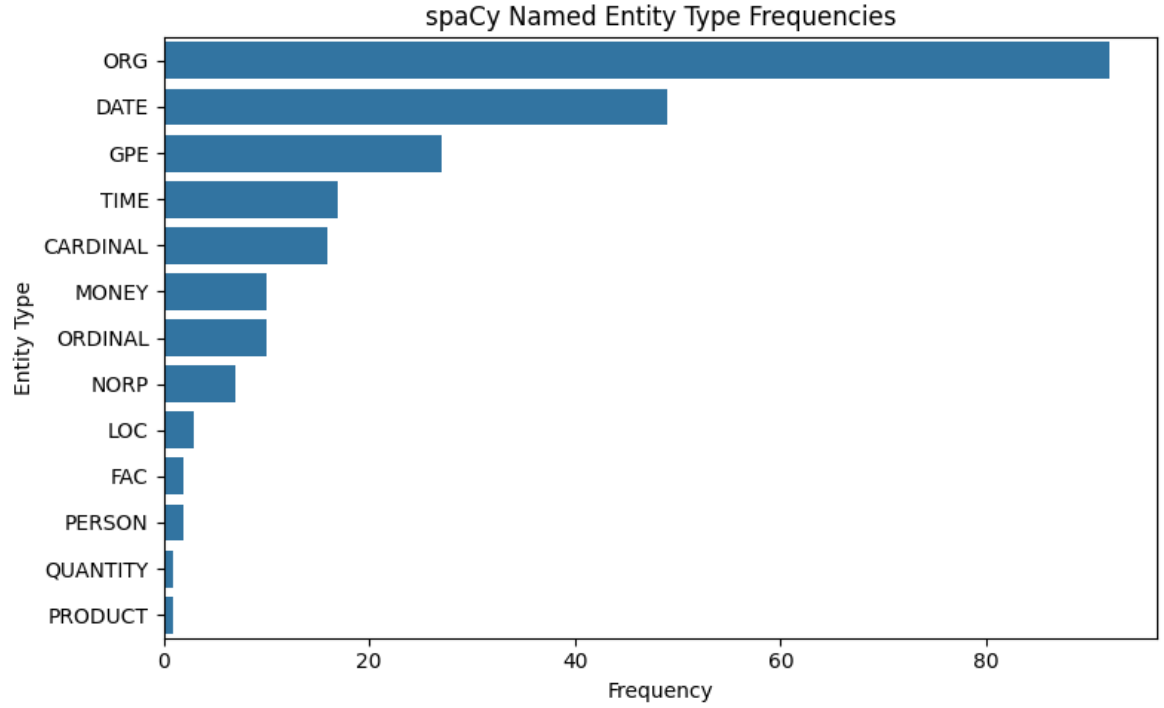


Figure 3: Entity Type Frequency Chart

(Source: self-developed)

Reflection on Preprocessing Choices

Preprocessing is crucial to a chatbot, as mentioned in my project reflection. Lemmatisation can be used to counteract sparse vocabulary by collapsing synonyms of words into bases and enhancing intent detection. Named Entity Recognition provides more insights into the bot's understanding, such as money, dates, places and orgs. The trial without stop words indicated that the number of words stored in the vocabulary rose to 760, the additional words being 'I,' 'my, and 'to. The removal of stopwords identified more bank-specific terms and could potentially improve the functioning of the model. However, some small words help to clarify intent, as well. Summarising the general results, the spaCy

method was more focused and precise to understand user's input in terms of language, which is appropriate when it comes to applications that need a chatty interaction with AI (Santos and da Silva, 2023).

Task 2: Classical NLP Baseline for Intent Classification

TF-IDF Feature Engineering

For creating a baseline system to classify intents in banking chatbots, the team used the Banking77 dataset. They converted this data set into numbers using TF-IDF. They removed the English stop words and added unigrams, bigrams and capped the max vocab at 10k in this process. As a result, the training matrix ended up being 9,993 by 10,000, and the test matrix was 3,076 by 10,000. They picked TF-IDF since it does a good job of highlighting important words and phrases while minimising the impact of common yet uninformative terms (Madatov, Sattarova and Vičič, 2025).

```
# Initialize a TF-IDF vectorizer to convert text into numerical feature representations.
tfidf = TfidfVectorizer(
    lowercase=True,
    stop_words="english",
    ngram_range=(1, 2),
    max_features=10000
)

# Learn the vocabulary from the training data and transform the texts into TF-IDF vectors.
X_train_tfidf = tfidf.fit_transform(X_train)

# Transform the test data using the vocabulary learned from the training set.
X_test_tfidf = tfidf.transform(X_test)

# Display the dimensions of the resulting training feature matrix.
print("TF-IDF training matrix shape:", X_train_tfidf.shape)

# Display the dimensions of the resulting test feature matrix.
print("TF-IDF test matrix shape:", X_test_tfidf.shape)
```

```
TF-IDF training matrix shape: (9993, 10000)
TF-IDF test matrix shape: (3076, 10000)
```

Figure 4: TF-IDF Feature Matrix Output

(Source: self-developed)

Classical Model Comparison

Three classical machine learning algorithms - Logistic Regression, Multinomial Naive Bayes and Linear Support Vector Machine (SVM) were tested. Linear SVM achieved the best accuracy (84.27%), weighted precision (84.75%), recall (84.27%) and F1 score (84.22%). Logistic Regression gave the best F1 score of 83.38 %, and Naive Bayes was the worst with an F1 score of 76.16 %.

The other problem was that Naive Bayes was very fast at training (0.03 seconds), but the assumption of independence of the features was problematic when the features are often correlated, like they are in banking data. Linear SVM classifier gave the fastest speed and accuracy balance and thus was used as the baseline classifier (Palanivinayagam, El-Bayeh and Damaševičius, 2023).

```
# Display the comparative performance results.
display(classical_results)
```

	Model	Accuracy	Precision	Recall	F1	Training Time Seconds
2	Linear SVM	0.842653	0.847469	0.842653	0.842225	0.980195
0	Logistic Regression	0.833875	0.841597	0.833875	0.833788	12.589548
1	Naive Bayes	0.770481	0.811897	0.770481	0.761621	0.034379

Figure 5: Classical Model Comparison Table

(Source: self-developed)

Confusion Matrix and Error Analysis

The confusion matrix showed high diagonal dominance, meaning that most categories were misclassified correctly, and that TF-IDF features were able to capture the banking vocabulary of each category (Markoulidakis and Markoulidakis, 2024). But there was some confusion between semantically similar intents, specifically about the why_verify_identity and verify_my_identity intents, and the pending_transfer and failed_transfer intents. In some cases, the prediction was incorrect, e.g., How to verify ID? was predicted as unable_to_verify_identity, whereas When will my transfer go through? was predicted as failed_transfer. These errors indicate that the classifier was mainly using common words, and the context meaning was not fully understood (Bercaru *et al.*, 2023).

	True Intent	Predicted Intent	Count
315	why_verify_identity	verify_my_identity	11
194	pending_transfer	failed_transfer	6
168	order_physical_card	getting_spare_card	6
24	card_acceptance	card_not_working	5
299	unable_to_verify_identity	verify_my_identity	5
310	virtual_card_not_working	get_disposable_virtual_card	5
140	fiat_currency_support	exchange_via_app	5
302	verify_my_identity	unable_to_verify_identity	5
223	top_up_by_bank_transfer_charge	transfer_fee_charged	5
142	fiat_currency_support	supported_cards_and_currencies	4
198	pending_transfer	transfer_timing	4
14	balance_not_updated_after_cheque_or_cash_deposit	top_up_by_cash_or_cheque	4
311	virtual_card_not_working	getting_virtual_card	4
50	card_payment_fee_charged	top_up_by_card_charge	3
221	top_up_by_bank_transfer_charge	receiving_money	3

Figure 6: Most common intent confusions

(Source: self-developed)

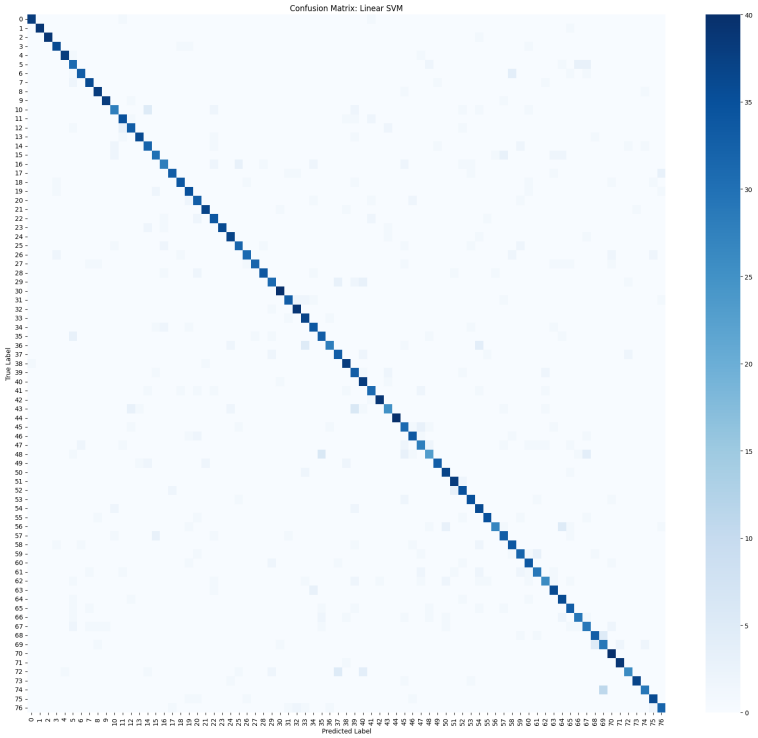


Figure 7: Linear SVM Confusion Matrix

(Source: self-developed)

```
display(
    misclassified[
        ["text", "intent", "predicted_intent"]
    ].sample(10, random_state=SEED)
)
```

	text	intent	predicted_intent
2618	how does a virtual card work	get_disposable_virtual_card	virtual_card_not_working
3010	What's the process for ID verification?	verify_my_identity	unable_to_verify_identity
1105	I have an unauthorized transaction on my state...	card_payment_not_recognised	extra_charge_on_statement
2757	whats your exchange rate	exchange_charge	exchange_rate
1859	When will my transfer go through?	pending_transfer	failed_transfer
977	I am traveling to Germany, Will I be able to u...	card_acceptance	card_not_working
2479	Where can I get my card at?	order_physical_card	getting_spare_card
2496	How much does it cost in fees to use your card?	order_physical_card	top_up_by_card_charge
3033	What all am I required to show for the identit...	verify_my_identity	why_verify_identity
1384	Can you explain the virtual cards limit?	disposable_card_limits	getting_virtual_card

Figure 8: Misclassified Examples Table

(Source: self-developed)

NER Feature Enhancement and Chatbot Implications

To find out if including entity info will increase classification accuracy, spaCy NER features with TF-IDF reps have been combined. The base model had an F1 score of 82.75% while the Linear SVM model performed slightly better with an F1 score of 84.61%. Named entities do provide additional context, but it is not as informative as the lexical properties in this dataset (Chen and Yin, 2022).

```
# Display the evaluation results for the combined-feature Linear SVM model.
print("TF-IDF + NER Linear SVM Results")
print("Accuracy:", ner_svm_acc)
print("Precision:", ner_svm_precision)
print("Recall:", ner_svm_recall)
print("F1:", ner_svm_f1)
print("Training time:", ner_svm_time)
```

```
TF-IDF + NER Linear SVM Results
Accuracy: 0.846553966189857
Precision: 0.8520526055439204
Recall: 0.846553966189857
F1: 0.8460860666531536
Training time: 0.7962546348571777
```

Figure 9: TF-IDF + NER Results

(Source: self-developed)

This kind of classifier may be applicable to a real-life chatbot to efficiently categorise regular customer requests to the proper department. It is a problem, however, that it relies a lot on the words that are actually present in the text, and therefore is quite weak with more complex demands, which contain more vague words or are based on a more complex context. Hence, in the case of real chatbots, it is necessary to have features such as confidence scores, follow-up questions and contingencies to ensure smooth operations (Chen and Yin, 2022).

Task 3: Semantic Embeddings for a Retrieval-Based Chatbot

Word2Vec Training

Banking77 corpus was preprocessed, and a Word2Vec model was trained on it using the CBOW architecture (sg=0), vector size=100, window size=5, minimum word frequency=2. The model was trained to use 926 distinct words. Word2Vec, in contrast to TF-IDF, is able to model the semantic relationships by embedding words as dense vectors, which causes the words with similar meaning or usage contexts to be closer together in the word embedding space (Mitera-Kiełbasa and Zima, 2024).

Semantic Similarity Analysis

The cosine similarity analysis showed weak semantic similarity between the banking-related terms. For instance, the similarity score of account-bank was 0.9986, and those of cash-withdraw and transfer-payment were 0.9946 and 0.9943, respectively. Meaningful clustering was also validated by nearest-neighbour analysis: card was grouped with terms like credit, order, freeze and payment with terms like transaction, statement, dispute. The results show that Word2Vec can accurately capture the domain-specific semantics, demonstrating its applicability in retrieval-based chatbot systems and semantic query matching (Goh *et al.*, 2023; Gao, Liu and Zhang, 2025).

```
# Display the semantic similarity scores for the selected word pairs.
display(similarity_df)
```

	Word 1	Word 2	Cosine Similarity
0	card	payment	0.993820
1	card	cash	0.984180
2	transfer	payment	0.994256
3	account	bank	0.998577
4	cash	withdraw	0.994560

Figure 10: Cosine Similarity Table

(Source: self-developed)

Embedding Visualisation

The research on the embedding visualisation is explored in the next part. Principal Component Analysis has been added to obtain 100-dimensional embeddings from two dimensions. A few clusters of banking concepts were interestingly presented by the PCA. Payment, transfer, card, and banking services words are grouped together according to their meanings. This clustering is justified by the fact that Word2Vec has grouped the vocabulary by the context where they are used in the text (Kitao, 2022).

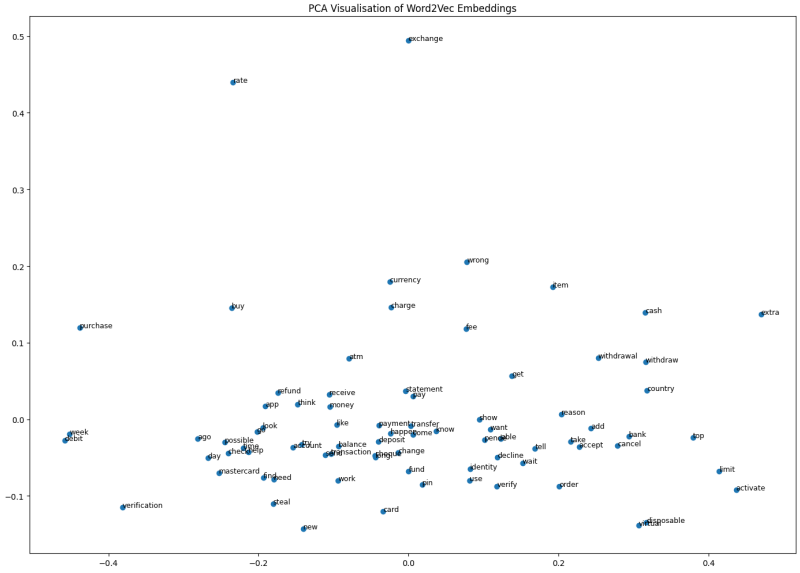


Figure 11: PCA Visualisation of Word2Vec Embeddings

(Source: self-developed)

Retrieval-Based Chatbot Demonstration

However, by generating sentence embeddings using the Word2Vec vectors and constructing a retrieval index with 3,000 customer queries, a retrieval-based chatbot was created. The most relevant responses were then identified using cosine similarity (Pandey and Sharma, 2023). For the query "I have not received my new bank card yet", the system retrieved many highly relevant examples with similar scores mentioned above, and the main examples are related to the card_arrival intent. This shows the usefulness of Word2Vec embeddings in semantic query matching and retrieval of responses.

User query: I have not received my new bank card yet

	text	intent	similarity
2782	Why haven't I received my new card yet?	card_arrival	0.999886
278	Can you tell me why I haven't received my new ...	card_arrival	0.999885
1500	Where can I have a new bank card sent to China?	card_about_to_expire	0.999883
2451	How long should it take for my new card to arr...	card_arrival	0.999867
1974	How do I know when my new card is going to arr...	card_arrival	0.999824

Figure 12: Retrieval Results Table

(Source: self-developed)

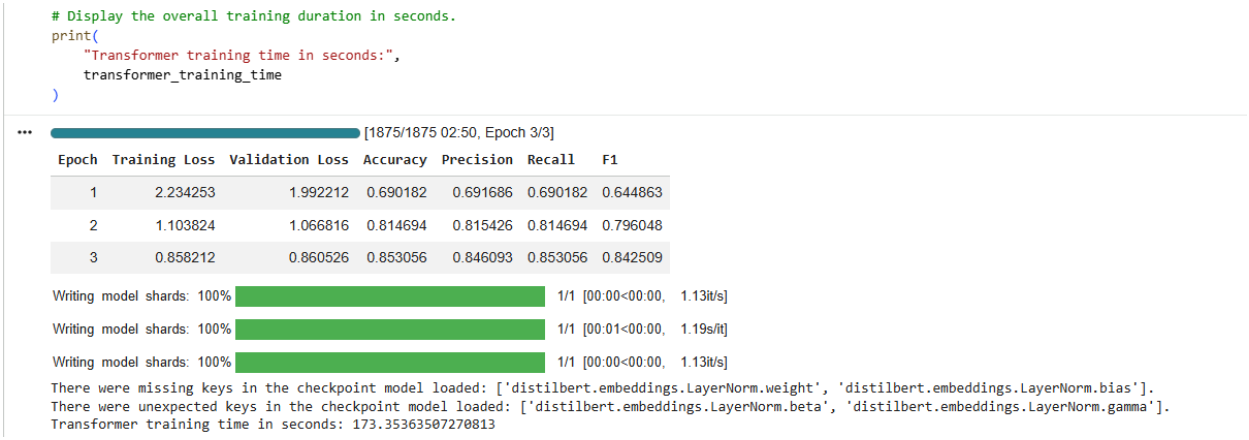
Implications for Chatbot Design

Word embeddings enhance retrieval-based chatbots by allowing them to match user queries with pre-existing responses, even if the words are altered. Cosine similarity is used to rank candidate responses and provide the most relevant responses. They also facilitate paraphrase handling by locating semantically related expressions next to each other in the embedding space. The retrieval experiment demonstrated the effectiveness of Word2Vec to match banking-related queries. It is because Word2Vec embeds a single vector per word that it might fail in cases where words have multiple meanings, or words do not appear in the training data set, or words have different meanings in different contexts, which would reduce its capabilities in more sophisticated conversational situations (Lee and Rew, 2025).

Task 4: Fine-Tuning a Transformer for Chatbot Intent Recognition

Dataset Preparation and Fine-Tuning

DistilBERT was chosen for intent recognition because it is efficient and has a great contextual understanding. The Banking77 dataset was tokenised using the DistilBERT tokeniser with a maximum sequence length of 64 tokens, and dynamic padding was employed to ensure that the model trained in a way that used less memory (Lorenzoni *et al.*, 2024). The model was first trained on a large text corpus (distilbert-base-uncased) and then fine-tuned for 77 intent classes with a learning rate of 2e-5 for 3 epochs and a batch size of 16. The accuracy of the validation set increased from 69.0% to 85.3% after the 3rd epoch, showing a good learning of intent-specific patterns (Barbon and Akabane, 2022).



```
# Display the overall training duration in seconds.
print(
    "Transformer training time in seconds:",
    transformer_training_time
)
```

[1875/1875 02:50, Epoch 3/3]

Epoch	Training Loss	Validation Loss	Accuracy	Precision	Recall	F1
1	2.234253	1.992212	0.690182	0.691686	0.690182	0.644863
2	1.103824	1.066816	0.814694	0.815426	0.814694	0.796048
3	0.858212	0.860526	0.853056	0.846093	0.853056	0.842509

```
Writing model shards: 100% 1/1 [00:00<00:00,  1.13it/s]
Writing model shards: 100% 1/1 [00:01<00:00,  1.19s/it]
Writing model shards: 100% 1/1 [00:00<00:00,  1.13it/s]
There were missing keys in the checkpoint model loaded: ['distilbert.embeddings.LayerNorm.weight', 'distilbert.embeddings.LayerNorm.bias'].
There were unexpected keys in the checkpoint model loaded: ['distilbert.embeddings.LayerNorm.beta', 'distilbert.embeddings.LayerNorm.gamma'].
Transformer training time in seconds: 173.35363507270813
```

Figure 13: DistilBERT Training Logs

(Source: self-developed)

Transformer Performance Evaluation

The final DistilBERT model got an accuracy of 85.31%, with a weighted precision of 84.61%, recall of 85.31%, and F1-score of 84.25%. This is a pretty good representation of the 77 banking categories, all with good intent classification.

The intents that had great scores in the classification report are age_limit, apple_pay_or_google_pay, passcode_forgotten and verify_top_up, which are all performing close to perfectly. The other way around, some intents were extremely difficult, such as card_acceptance, virtual_card_not_working, and why_verify_identity, where there was a common vocabulary between them (Nogueira *et al.*, 2022).

reverted_card_payment?	0.78	0.88	0.82	40
supported_cards_and_currencies	0.76	0.78	0.77	40
terminate_account	0.83	1.00	0.91	40
top_up_by_bank_transfer_charge	0.94	0.78	0.85	40
top_up_by_card_charge	0.93	1.00	0.96	40
top_up_by_cash_or_cheque	0.97	0.85	0.91	40
top_up_failed	0.84	0.93	0.88	40
top_up_limits	0.90	0.95	0.93	40
top_up_reverted	0.79	0.85	0.82	40
topping_up_by_card	0.83	0.47	0.60	40
transaction_charged_twice	0.89	1.00	0.94	40
transfer_fee_charged	0.88	0.93	0.90	40
transfer_into_account	0.84	0.80	0.82	40
transfer_not_received_by_recipient	0.92	0.85	0.88	39
transfer_timing	0.70	0.88	0.78	40
unable_to_verify_identity	0.85	0.72	0.78	40
verify_my_identity	0.62	0.82	0.71	40
verify_source_of_funds	0.95	1.00	0.98	40
verify_top_up	1.00	1.00	1.00	40
virtual_card_not_working	0.00	0.00	0.00	40
visa_or_mastercard	0.97	0.93	0.95	40
why_verify_identity	0.55	0.45	0.49	40
wrong_amount_of_cash_received	0.92	0.88	0.90	40
wrong_exchange_rate_for_cash_withdrawal	1.00	0.72	0.84	40
accuracy			0.85	3076
macro avg	0.85	0.85	0.84	3076
weighted avg	0.85	0.85	0.84	3076

Figure 14: Classification Report

(Source: self-developed)

The best-performing model was DistilBERT with an accuracy of 85.31%, which outperformed the other two models, Linear SVM and Logistic Regression. DistilBERT achieved the highest accuracy, whereas the TF-IDF + NER Linear SVM achieved a marginally higher F1-score (84.61% compared with 84.25%). This suggests that transformer models do not always outperform carefully engineered classical approaches on structured intent-classification datasets. While DistilBERT benefited from contextual language understanding, the combination of TF-IDF and NER features was highly effective at capturing discriminative intent signals within the Banking77 dataset. However, there is one problem - they're much more expensive when it comes to computing resources. DistilBERT required a training time of roughly 173 seconds, as compared to less than 13 seconds that was required by the traditional models. So, the models receive high marks in tests, but need to make a higher demand on the CPU and time.

TF-IDF + NER Linear SVM performed better with an F1 of 84.61% while DistilBERT had an F1 of 84.25%. This suggests that when they are well engineered, traditional features are effective in structured intent classification tasks.

```
# Display the final comparative evaluation table.
display(comparison_table)
```

	Model	Accuracy	F1 Score	Training Time Seconds	Model Complexity	Chatbot Deployment Comment
3	Linear SVM + NER	0.846554	0.846086	0.796255	Medium	Tests whether entity signals improve intent de...
4	DistilBERT	0.853056	0.842509	173.353635	High	Best semantic understanding but most expensive
2	Linear SVM	0.842653	0.842225	0.980195	Medium	Strong classical baseline for sparse text
1	Logistic Regression	0.833875	0.833788	12.589548	Medium	Interpretable and efficient baseline
0	Naive Bayes	0.770481	0.761621	0.034379	Low	Very fast but weaker with overlapping intents

Figure 15: Classical vs Transformer Comparison Table

(Source: self-developed)

Transformer NER and Chatbot Applications

In another experiment, spaCy's Named Entity Recognition (NER) model was compared to the transformer-based model dslim/bert-base-NER. The transformer was more successful at identifying specific names, including 'John', 'London' and 'Barclays'. Sometimes, though, spaCy was too general, seeing the phrase "John yesterday" as a single entity instead of a single name "John".

This is a demonstration of how accurate entity extraction results can be obtained with NER based on a transformer. Accurate entity recognition is very useful in the task of slot filling, personalising, transaction tracking, and even generating dynamic dialogues in chatbots (Tao *et al.*, 2022).

Text: I transferred 500 pounds to John yesterday but it has not arrived.

Transformer NER:

	entity_group	score	word	start	end
0	PER	0.997898	John	28	32

spaCy NER:

	Entity	Label
0	500 pounds	QUANTITY
1	John yesterday	PERSON

Text: My Visa card was declined in London.

Transformer NER:

	entity_group	score	word	start	end
0	MISC	0.973048	Visa	3	7
1	LOC	0.999545	London	29	35

spaCy NER:

	Entity	Label
0	Visa	ORG
1	London	GPE

Text: I want to open a new account with Barclays.

Transformer NER:

	entity_group	score	word	start	end
0	ORG	0.999165	Barclays	34	42

spaCy NER:

	Entity	Label
0	Barclays	ORG

Figure 16: spaCy vs Transformer NER Comparison

(Source: self-developed)

Discussion and Reflection

DistilBERT's intent recognition is most effective because it can better grasp the context of sentences than the keyword-based models. The majority of errors took place between related intents like "identity verification" and "transfer". The results demonstrate that the training data plays an important role, as it led to a drop in accuracy from 85.3% to 51.0% when the amount of training data was halved. Transformer models provide more context-aware understanding and classification accuracy, but they also

have higher computational, memory, and training time demands, requiring a balance between performance, efficiency, and real-world use, especially for chatbots.

Task 5: Responsible Conversational AI: Risks and Mitigation

Ethical and Safety Risks

Safety, reliability, fairness, and accuracy should be key aspects of the intent recognition of a chatbot. In banking, where it can cause a tremendous problem should they make a mistake on a sensitive issue, say card problems or the loss of money. Consumers don't want to overlook critical steps or lose money, particularly when it affects critical factors like trust in an industry such as financial services, where bots are playing an increasingly important role (Lin, Huang and Yang, 2023).

Observed Failure Cases

The findings of error analysis showed that the two models (classical and transformer) had experienced difficulties in distinguishing between similar intents. For example, they swapped 'why_verify_identity' with 'verify_my_identity' as these intents have similar words, and are relatively close in meaning. TF-IDF + Linear SVM was also found to be problematic since this model excessively depended on word-level information, that is hard to capture context information.

Although DistilBERT performed better on the whole, it showed its downfalls on certain intents, such as 'card_acceptance' and 'virtual_card_not_working'. The accuracy was greatly reduced for these classes. All of these generate a significant worry for reliability and fairness - if these are mishandled by some customers' requests, it can lead to inconsistent and poor answers from the Chatbot (Raza *et al.*, 2025).

```
Confidence threshold: 0.6
Number of low confidence cases: 1531
Percentage of test set: 49.77 %
```

	text	intent	predicted_intent	confidence	correct
0	How do I locate my card?	card_arrival	card_arrival	0.375907	True
2	I ordered a card but it has not arrived. Help ...	card_arrival	card_arrival	0.470986	True
3	Is there a way to know when my card will arrive?	card_arrival	card_arrival	0.358120	True
5	When will I get my card?	card_arrival	card_delivery_estimate	0.494216	False
9	Is it normal to have to wait over a week for m...	card_arrival	card_arrival	0.566454	True
11	How long does a card delivery take?	card_arrival	card_delivery_estimate	0.439474	False
15	I have been waiting longer than expected for m...	card_arrival	card_arrival	0.437666	True
16	I've been waiting longer than expected for my ...	card_arrival	card_arrival	0.549642	True
20	I did not get my card yet, is it lost?	card_arrival	card_arrival	0.576045	True
21	Status of the card I ordered.	card_arrival	card_arrival	0.412175	True
22	How long should my new card take to arrive?	card_arrival	card_arrival	0.397144	True
32	How do I know when my card will arrive?	card_arrival	card_delivery_estimate	0.339991	False
33	I'm still waiting on my card to be delivered.	card_arrival	card_arrival	0.515499	True
36	I'm starting to think my card is lost because ...	card_arrival	card_arrival	0.470100	True
37	Is there tracking info available?	card_arrival	unable_to_verify_identity	0.173568	False
40	Why won't my card show up on the app?	card_linking	card_linking	0.540353	True
41	I would like to reactivate my card.	card_linking	card_linking	0.582616	True
45	I received my new card, but I don't see it in ...	card_linking	card_linking	0.515729	True
47	How do I add the card to my account?	card_linking	card_linking	0.551750	True
48	Can I put my old card back into the system? I ...	card_linking	card_linking	0.577138	True

Figure 17: Low-Confidence Transformer Predictions

(Source: self-developed)

	Intent	Number of Examples	Accuracy	Average Confidence
10	card_acceptance	40	0.000000	0.234500
72	virtual_card_not_working	40	0.000000	0.443514
74	why_verify_identity	40	0.450000	0.370346
62	topping_up_by_card	40	0.475000	0.223485
18	card_swallowed	40	0.550000	0.291901
22	compromised_card	40	0.575000	0.253753
37	get_disposable_virtual_card	40	0.675000	0.418094
27	declined_transfer	40	0.700000	0.571051
48	pending_transfer	39	0.717949	0.552481
68	unable_to_verify_identity	40	0.725000	0.362549
76	wrong_exchange_rate_for_cash_withdrawal	40	0.725000	0.537441
28	direct_debit_payment_not_recognised	40	0.750000	0.659019
5	balance_not_updated_after_bank_transfer	40	0.750000	0.511259
56	top_up_by_bank_transfer_charge	40	0.775000	0.423671
12	card_delivery_estimate	40	0.775000	0.416500

Figure 18: Per-Intent Weakness Analysis

(Source: self-developed)

Confidence, RAG, Guardrails, and Mitigation Strategies

The top 0.60 threshold for the confidence score yielded 49.77% of the test set, or 1531 low confidence predictions. But the confidence scores were not high enough, even for several examples that are classified correctly, such as card_arrival and card_linking, which is not the same as prediction correctness. Thus, it is important to differentiate between all predictions and ensure that the chatbots are not capable of differentiating between them. Whereas, in some scenarios with unclear information, asking for more user information and escalating to human agents can result in better prevention of inappropriate answers (Zafar *et al.*, 2025).

One recent development that is relevant to NLP is Retrieval-Augmented Generation (RAG). This helps minimise the risk of incorrect or fabricated information, especially in critical banking transactions, where it comes to payment, fees, card services, and identity verification. Furthermore, guardrails act as safety mechanisms by filtering out dangerous outputs and passing on complicated cases to human review (Swacha and Gracel, 2025).

The aforementioned weaknesses identified in Tasks 2-4 are mitigated through the following recommendations: Data Balancing: Place a threshold on the quantity and variety of data that the Chatbot can access. Underperforming Intent Augmentations: Set thresholds for the quality of intent data that the Chatbot uses.Confidence Threshold: Introduce a limit on the confidence level at which the Chatbot responds to user queries.

Integrated Reflection on Performance and Design Choices

The experimental results indicate that each NLP technique was used in different parts of the chatbot pipeline. spaCy preprocessing yielded a smaller vocabulary and reduced the noise compared with NLTK preprocessing prior to modelling, due to the regularised nature of the vocabulary. For short and structured intent classification, the traditional machine learning methods still work well, with TF-IDF + Linear SVM achieving a good baseline accuracy of 84.27%. The SVM F1 score can be improved slightly by incorporating the Named Entity Recognition features, with an F1 score of 84.61%. Entity information also provides some context and value. Through the effective matching of queries, Word2Vec embeddings were able to describe semantic relationships and enable the ability to retrieve a chatbot. DistilBERT achieved the highest overall performance with 85.31% accuracy. However, the improvement required a lot more computational resources and training time, and at times, similar intents, such as identity verification, were confused with card-related intents.

Conclusion

An end-to-end NLP pipeline for banking chatbots has been developed in this project on the Banking77 dataset. It contained responsible AI analysis, traditional ML, semantic embeddings and preprocessing. It is found that the Linear SVM with TFIDF performed well as a baseline, and DistilBERT obtained the best accuracy because of its contextual understanding. Word2Vec also facilitated the retrieval-based matching.

One of the most important takeaways is that whilst correctness is key, it is not an absolute requirement for a safe Chatbot deployment. Some intents, such as verification and identity, card issues, and transfer status, were hard to define with both classical and transformer models, showing up in the error analysis. But, as the confidence analysis showed, some lots were not very confident.

The disadvantages are limited training, no actual customer interactions to test the transformers and only trained on the English Banking77 dataset. Larger contextual models, Retrieval-Augmented Generative models, multilingual models/datasets, more data augmentation when intents are underrepresented, and human oversight for escalation scenarios will be studied for more reliable, safer results in future work.

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